

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

SCSR2213 NETWORK COMMUNICATIONS

Task#4

Group 4: Linux

SECTION: 5

Lecturer: Dr. Adnan Khalid

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In the previous tasks of our project, we have established a solid foundation for the development of the new Faculty of Computing building. In Task 1, we formed our team, "Linux" and collaborated to design a comprehensive floor plan for the two-story building. The layout includes classrooms, laboratories, and common areas, ensuring that it accommodates the needs of both students and faculty. Additionally, we documented our initial discussions in meeting minutes, which helped us track our progress and decisions.

In Task 2, we carried out a preliminary analysis to gather relevant information about the project. We formulated over 10 focused questions to clarify our requirements and gain deeper insights. Through research and consultations with stakeholders, we identified both the current and potential future needs of the building. This allowed us to assess the feasibility of the project and refine our design plan to meet the expectations of all involved parties.

Task 3 focused on selecting the appropriate Local Area Network (LAN) devices that would meet the objectives outlined in our case study. Our main goal was to ensure that the new building would have an efficient and secure network infrastructure to support both current and future demands. We evaluated various options to determine the most suitable devices for our needs.

In Task 4, we designed the physical network infrastructure of the building. This task involved deciding where each network device would be located, identifying the best cabling routes, and establishing connections between floors. We selected appropriate media, including twisted pair cables, fiber optic cables, and wireless technology, to ensure the network's reliability and efficiency. This careful planning is essential to guarantee smooth communication across all devices within the building.

Today, in Task 5, we will focus on subnetting. This task involves assigning unique IP addresses to each device in the different rooms of the building to avoid any conflicts. Using the provided network address, we will divide the larger network into smaller, more manageable subnets tailored to each lab and room. This structured approach will help us use IP addresses efficiently and ensure that the network is well-organized and free of overlaps.

These tasks have progressively built upon one another, taking us through each crucial stage of the project. With each completed task, we have gained a clearer understanding of the requirements and challenges, bringing us closer to successfully completing the Faculty of Computing building.

Identifying the Network address:

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1-Network Subnetting:

Based on the provided network address of 192.17.0.0/8, we decided to divide the subnet according to the number of devices required for each working area. The subnetting was carried out as follows:

Subnet No	Area	CIDR Subnet	IP Range	No of Usable Addresses
0	Server Room & Hybrid class room and video conference room	192.17.1.0/27	192.17.1.0 - 192.17.1.31	30
1	Lecturer's office	192.17.1.32/27	192.17.1.32 - 192.17.1.63	30
2	academic office & meeting room	192.17.1.64/26	192.17.1.64 - 192.17.1.127	62
3	General Lab 1 & 2	192.17.1.128/25	192.17.1.128 – 192.17.1.255	128
4	Cisco Network Lab & Embedded Lab and student lounge	192.17.2.0/25	192.17.2.0 – 192.17.2.127	128

IP Assignation to working areas:

Ground Floor

1. Server Room

Devices	IP Address
Default Gateway	192.17.1.1
Primary Router	192.17.1.2
Backup Router	192.17.1.3
Primary Firewall	192.17.1.4
Backup Firewall	192.17.1.5
Core Switch	192.17.1.6
SV1	192.17.1.7
SV2	192.17.1.8
SV3	192.17.1.9
SV4	192.17.1.10
PC1	192.17.1.11
PC2	192.17.1.12

2. Hybrid Classroom

Devices	IP Address
Default Gateway (same as server room)	192.17.1.1
PC3	192.17.1.13
Smart TV1	192.17.1.14
Smart TV2	192.17.1.15

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3. Video Conferencing Room

Devices	IP Address
Default Gateway (same as server room)	192.17.1.1
Smart TV3	192.17.1.16

4. Lecturer's office

Devices	IP Address
Default Gateway	192.17.1.33
PC4	192.17.1.34
PC5	192.17.1.35
PC6	192.17.1.36
…	…
PC21	192.17.1.51
WAP1	192.17.1.52

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5. Academic Office

Devices	IP Address
Default Gateway	192.17.1.65
PC22	192.17.1.66
PC23	192.17.1.67
PC24	192.17.1.68
…	…
PC48	192.17.1.92
Printer1	192.17.1.93
Printer2	192.17.1.94
Printer3	192.17.1.95
Copier1	192.17.1.96
Copier2	192.17.1.97
Smart TV4	192.17.1.98
Smart TV5	192.17.1.99
WAP2	192.17.1.100

6. **Meeting Room**

Devices	IP Address
Default Gateway (same as the Academic Office)	192.17.1.65
Smart TV6	192.17.1.101

First Floor:

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7. **General Lab 1**

Devices	IP Address
Default Gateway	192.17.1.129
PC49	192.17.1.130
PC50	192.17.1.131
PC51	192.17.1.132
…	…
PC79	192.17.1.160
Smart TV7	192.17.1.161
Switch 1	192.17.1.162
Switch 2	192.17.1.163

8. **General Lab 2**

Devices	IP Address
Default Gateway (same as general lab 1)	192.17.1.129
PC80	192.17.1.164
PC81	192.17.1.165
PC82	192.17.1.166
…	…
PC110	192.17.1.194
Smart TV8	192.17.1.195
Switch 3	192.17.1.196
Switch 4	192.17.1.197

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9. Cisco Network Lab

Devices	IP Address
Default Gateway	192.17.2.1
PC111	192.17.2.2
PC112	192.17.2.3
PC113	192.17.2.4
…	…
PC141	192.17.2.32
Smart TV9	192.17.2.33
Switch 5	192.17.2.34
Switch 6	192.17.2.35

10. Embedded Lab

Devices	IP Address
Default Gateway (same as cisco network lab)	192.17.2.1
PC142	192.17.2.36
PC143	192.17.2.37
PC144	192.17.2.38
…	…
PC172	192.17.2.66
Smart TV10	192.17.2.67
Switch 7	192.17.2.68
Switch 8	192.17.2.69

11. Student lounge

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Devices	IP Address
Default Gateway (same as cisco network lab)	192.17.2.1
Smart TV 11	192.17.2.70
Copier 3	192.17.2.71

3-Meeting Minutes:

Meeting Minutes

Date/Time	18 Jun 2025 10:00 pm		
Location	Online (google meet)		
Agenda	Distribution of IP address		
Meeting MC	Garrah Thabit Mohammed		
Attendance			
Name		Time	Reason For Absence
Garrah Thabit Mohammed		10:00 pm	-
MOHAMMED ALSAKKAF		10:00 pm	-
LEE TUNG ERN		10:15 pm	-
Mohammed Abdulrahim		10:10 pm	-
Minutes			
No.	Items Discussed	Ideas/Suggestions and Person Giving It	Person in Charge & Date

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1.	Network subnetting	Garrah completed the network subnetting and presented it to the team. After reviewing it, everyone agreed that it was well done and accepted the proposed solution.	Garrah
2.	The Ip address of the devices in work areas	ALSAKKAF proposed creating a table to list all the IP addresses of the devices.	ALSAKKAF
		took the responsibility of assigning the IP addresses for the work areas on the ground floor.	The Team
		took the responsibility of assigning the IP addresses for the work areas on the first floor.	The Team
3.	Meeting ended	Meeting ended at 11.30 pm.	-

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